

Assignment - 7

Title : Expression Evaluation using stack.

Problem Statement :

Write a C program to evaluate postfix expression using stack.

Objective

- 1) To study stack and its operations.
- 2) To study the importance of expression evaluation.

Theory :

→ Expression evaluation is the process of calculating the value of an arithmetic expression. A stack is the most suitable data structure because it follows the last in first out (LIFO) principle, which matches the order in which operators and operands are processed.

Expression evaluation is commonly performed on:

- Postfix (Reverse Polish Notation)
- Prefix (Polish Notation)

Infix expressions are usually converted to postfix or prefix before evaluation.

Ex:

Evaluate the expression $35 + 4^*$

Symbol	operation	Stack
3	push 3	3
5	push 5	5, 3
+	Pop 5, Pop 3, perform $5+3$, push 8	8
4	push 4	4, 8
*	Pop 4, Pop 8, perform $4*8$, push 32	32

Implementation:

Platform:

- 64 bit open source linux or its derivatives
- open source C programming tool like gcc / Eclipse Editor

Test conditions:

- 1) Input postfix expression
- 2) evaluate the value of expression

Pseudocode

→ Postfix Evaluation

Postfix_EVALUATION (expression)

Create an empty stack

For each symbol in expression

If symbol is an operand
push (symbol)

Else if symbol is an operator

operand 2 = pop()

operand 1 = pop()

result = operand 1 operator operand 2
push(result)

answer = pop()

Return answer.

Time Complexity

→ For an expression containing n symbols
operations.

push

$O(1)$

pop

$O(1)$

$O(n)$

Postfix Evaluation

$O(n)$

Space Complexity

FAQs

1) Explain how postfix expression is evaluated.

Ans → In postfix, evaluation, operands are pushed into the stack. whenever an operator is encountered, the top two operands are popped, the operation is performed, and the result is pushed back into the stack.

For Ex:

$$35 + 4*$$

Steps:

3 → Push

5 → Push

* → S and 5 popped.

2 FS are popped
Push 8.

4 → Push

* $\rightarrow 4$ and 8 are popped

$$8 \times 4 = 32$$

Push 32

2 Explain how prefix expression is evaluated.

$\rightarrow$ In prefix notation, the operation comes before the operands.

$$E^x: x + 357$$

$\Rightarrow$ To evaluate prefix, scan the expression from right to left.

4 $\rightarrow$ Push

5 $\rightarrow$ Push

3 $\rightarrow$ Push

$$+ \rightarrow 3 + 5 = 8$$

$$* \rightarrow 8 * 7 = 32$$

3) What is ISP and ICP?

Ans ISP means In-Stack Priority.

ICP means Incoming Priority

They are mainly used when handling operators such as

$+, -, *, /$ and $\wedge$ especially during infix to postfix / prefix conversion.

4) Give various example stack!

Ans Acc to ~~books~~

1) Expression evaluation

2) Infix to postfix conversion

3) postfix to prefix conversion

4) Function call management.

5) Recursion

6) Parenthesis matching

7) Undo/redo operations

8) Browser back / Forward queries

9) Syntax parsing.

10) String reversal.

• Conclusion:

→ Thus, implemented stack operations to evaluate the value of an expression.